

Assignment 3 - Homework & Deliverables

PROJECT 1 | Team Members: Ava Owen, Qingyuan Yu,

Repository Link: <https://github.com/owena6-blip/EGR103-Project-1>

Coordinate with your team using your new GitHub repository to complete this document before the next class.

Grading Rubric (14 Points Total)

Tech Execution (6 Points):

- **2 pts:** GitHub Repository successfully created, synced, and used by the team.
- **2 pts:** Whiteboard photo properly embedded using Markdown syntax.
- **2 pts:** Mermaid.js code successfully generated, properly formatted in a code block, and renders a valid flowchart.

Mind Map & Concepts (8 Points):

- **4 pts:** Mind Map Breadth (Photo/Code shows robust brainstorming, integrating the blind stimulus with the 25kg challenge, minimum 15+ nodes).
- **4 pts:** Concept Evaluations (Each team member completed their pitch thoroughly, directly addressing how geometry manages the load, and accurately completed the constraint checks).

Section 1: Tech Setup & AI Digitization (6 Points)

1a. GitHub Student Pack (Do this immediately)

- Register for a free [GitHub Student Developer Pack](#). This gives you free access to GitHub Copilot. (*Approval can take time; do this today*).

1b. Whiteboard Photo

https://drive.google.com/file/d/10k7dFNnjD0DqJKsb6ANF7Af1fUONnaHo/view?usp=drive_link

1c. AI-Generated Mermaid Code

- **Action:** Open **GitHub Copilot Chat** directly inside VS Code. Click the paperclip icon (or drag-and-drop) to upload the photo of your whiteboard.
- **Prompt:** "Act as an expert software engineer. Convert this hand-drawn mind map into valid Mermaid.js flowchart code. Use a top-down (TD) orientation. Ensure the concepts marked with stars are noted."
- **Fallback:** If Copilot struggles to read the handwriting, manually construct your map using

the visual editor at [MermaidChart.com](https://mermaidchart.com), then click "View Code" to copy it.

- **Result:** Paste the Mermaid code below and preview it in VSCode using your Markdown/Mermaid extension.

[Paste your code here]

Section 2: Rapid Individual Concept Evaluations (8 Points)

Each team member claims ONE of the starred concepts from the whiteboard to evaluate against the Phase 0/1 constraints. Ensure your pitches explain the mechanics of the geometric joints.

Concept 1 (Evaluator: _____)

- **Concept Name:** _____
- **The Pitch (1-2 sentences):** What is it and how does it manage the 25kg shifting load? (Use Copilot to refine your phrasing)
 - Answer: _____
- **Constraint Checklist:**
 - Cardboard ONLY (no glue/tape/metal)? [Y / N]
 - Operable by BOTH 1 AND 2 people? [Y / N]
 - What specific fastener-free geometry handles the connection? (e.g., interlocking tabs, friction-fit, precise folds): _____
 - What is the biggest risk for failure during the dynamic carry? (creasing, delamination, joint blowout): _____

Concept 2 (Evaluator: _____)

- **Concept Name:** _____
- **The Pitch (1-2 sentences):** What is it and how does it manage the 25kg shifting load?
 - Answer: _____
- **Constraint Checklist:**
 - Cardboard ONLY (no glue/tape/metal)? [Y / N]
 - Operable by BOTH 1 AND 2 people? [Y / N]
 - What specific fastener-free geometry handles the connection?:

 - What is the biggest risk for failure during the dynamic carry?:

Concept 3 (Evaluator: _____)

- **Concept Name:** _____
- **The Pitch (1-2 sentences):** What is it and how does it manage the 25kg shifting load?
 - Answer: _____
- **Constraint Checklist:**
 - Cardboard ONLY (no glue/tape/metal)? [Y / N]
 - Operable by BOTH 1 AND 2 people? [Y / N]
 - What specific fastener-free geometry handles the connection?:

 - What is the biggest risk for failure during the dynamic carry?:

Concept 4 (Evaluator: _____)

- **Concept Name:** _____
- **The Pitch (1-2 sentences):** What is it and how does it manage the 25kg shifting load?
 - Answer: _____
- **Constraint Checklist:**
 - Cardboard ONLY (no glue/tape/metal)? [Y / N]
 - Operable by BOTH 1 AND 2 people? [Y / N]
 - What specific fastener-free geometry handles the connection?:

 - What is the biggest risk for failure during the dynamic carry?:

Concept 5 (Evaluator: _____) *Only if 5-person team*

- **Concept Name:** _____
- **The Pitch (1-2 sentences):** What is it and how does it manage the 25kg shifting load?
 - Answer: _____
- **Constraint Checklist:**
 - Cardboard ONLY (no glue/tape/metal)? [Y / N]
 - Operable by BOTH 1 AND 2 people? [Y / N]
 - What specific fastener-free geometry handles the connection?:

 - What is the biggest risk for failure during the dynamic carry?:

Note: We will use these individual concept evaluations in our upcoming assignment to develop formal design criteria and perform a structured down-selection (using a Morphological Chart or Pugh Matrix). Ensure your GitHub commits are pushed and synced before the next class!